Measuring Financing Gaps with Machine Learning

Survey Evidence from SAFE in the European Union

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Abstract

This thesis measures firm-level financing gaps in the European Union using micro-data from the ECB's SAFE survey, framing the analysis as a missing-data problem in a large-scale survey environment. The discontinuation and item non-response of the loan-size question create a structural break in euro-valued gap measurement, requiring a robust strategy to recover systematically missing information. Alternative CatBoost-based machine-learning imputation models are developed and compared to reconstruct missing data points. Model selection follows an explicit decision rule that prioritises distributional credibility while imposing accuracy guardrails to ensure economically coherent aggregates. The preferred specification achieves high predictive accuracy and distributional alignment, revealing sizeable and persistent financing gaps across EU countries and continued financial fragmentation within the euro area. More broadly, the proposed framework is readily extendable to alternative imputation models and other survey environments.

Chapter 1

Introduction

Financing gaps are a central policy concern as they signal frictions in the allocation of credit to firms, with direct implications for investment, growth, and macroeconomic stability. When viable firms cannot obtain the external finance they seek, real economic activity may be constrained beyond what fundamentals justify. For policymakers, accurately measuring financing gaps is therefore essential to monitor credit conditions and design targeted interventions. Different methodologies have been proposed to estimate financing gaps, ranging from structural and model-based approaches to survey-based measures that directly capture firms' reported credit needs and outcomes. This thesis follows the survey-based approach using microdata from the ECB/European Commission Survey on the Access to Finance of Enterprises (SAFE), which provides harmonized cross-country information on firms' financing experiences.

The objective of this thesis is methodological. Rather than proposing a new substantive measure of financing gaps, it develops, compares, and validates alternative imputation models to address missing information in SAFE responses, with the goal of constructing a transparent and expandable decision framework for model selection. Missing data in key survey questions pose a non-trivial challenge for constructing firm-level financing gap measures. Because these gaps depend directly on reported demand and credit outcomes, any systematic bias in the imputation stage can mechanically distort aggregate estimates and policy conclusions.

This thesis, therefore, reframes the financing gap problem as a predictive and model-selection problem. Specifically, it evaluates alternative CatBoost-based machine learning specifications for imputing missing values. Two modelling strategies are constructed: a multi-class classification model that predicts bins directly, and a regression-based model that predicts bin midpoints and then maps them back to the discrete support. The imputation relies exclusively on SAFE survey variables, ensuring that the resulting dataset remains suitable for subsequent causal and policy analysis without mechanical overlap between the imputation stage and external explanatory covariates. The comparison is not limited to predictive accuracy in a narrow statistical sense, but instead embeds multiple evaluation spaces – bin space, euro space, and financing-gap space – to ensure that the preferred specification delivers economically coherent results. Model selection follows a pre-defined analytical decision rule that considers distributional credibility, accuracy, and predictive performance. Gap-space results are reported solely as robustness checks to ensure that the selected specification does not produce economically implausible financing-gap estimates.

Beyond choosing between two specifications and delivering a consistent firm-level dataset for policy-relevant analysis, the thesis contributes a generalizable framework for evaluating imputation models in policy-oriented survey contexts. By separating (i) model construction, (ii) multi-dimensional validation, and (iii) rule-based selection, the approach can be extended to alternative algorithms, additional survey waves, or other missing variables within SAFE and similar enterprise surveys. The ultimate contribution is therefore not a single preferred estimator, but a structured methodology that makes model choice transparent, reproducible, and expandable in future applications.

The remainder of the thesis first defines in Chapter 2 the financing-gap measures and data structure, then Chapter 3 presents the competing imputation models and their training design. Chapter 4 applies the decision framework to select the baseline specification. Chapter 5 provides robustness checks and descriptive evidence, and Chapter 6 concludes.

Chapter 2

Data and Construction of Financing-Gap Measures

This chapter describes the main data source used in the thesis and the empirical strategy adopted to construct firm-level financing-gap measures. The analysis follows a survey-based *"Need Method"* as described in the European Commission report (Mitra et al. 2026) which is here independently implemented and adapted using SAFE microdata.[1] First, firms' financing needs are inferred from responses on credit demand and results; second, these needs are translated into euro amounts using information on the requested loan size. The resulting firm-level gap measures are aggregated and compared across alternative definitions that differ in sample coverage. The treatment of missing loan-size information is discussed separately in the Chapter 3.

2.1 SAFE data

The analysis relies on the Survey on the Access to Finance of Enterprises (SAFE), a joint initiative of the European Commission and the European Central Bank launched in 2009 to

[1] Parts of the conceptual framework presented in this chapter draw on work conducted during the author's participation in a forthcoming European Commission report. The views expressed herein, as well as any errors, interpretations, or extensions, are solely those of the author.

monitor firms' financing conditions and needs. SAFE collects information on firms' use of external finance instruments (including bank loans), perceived constraints, and expectations, together with a set of firm characteristics. The survey dataset, as of today, contains approximately 430,000 observations across 37 waves, 42 countries, 4 sectors, and 4 size classes, and provides sampling weights designed to ensure representativeness at the country, size, and sector levels.[2] The financing-gap measures constructed in this thesis are based on the following SAFE questions, as already used by Mitra et al. (2026), and Raab (2025), namely:

- **Q7a, loan application status**: "Have you applied for the following types of financing during the previous and/or current quarter? a) Bank loan (excluding overdraft and credit lines); b) Trade credit; c) Other external financing; d) Credit line, bank overdraft or credit card borrowing."
- **Q7b, loan application outcome**: "If you applied and tried to negotiate for this type of financing during the previous and/or current quarter, what was the outcome?"
- **Q8a, loan size (ordinal brackets)**: "What is the size of the last bank loan that your enterprise obtained or renegotiated in the past 6 months? or attempted to obtain in the past 6 months?
- **Q32, discouragement / "shadow" demand**: "You mentioned that bank loans are not relevant for your enterprise. What is the main reason for this?

In addition, firm characteristics such as country (question $D0$), size (question $D1$), sector (question $D3$), and age (question $D5$) are used for descriptive breakdowns and subsequent empirical analysis, and are standard components of the SAFE microdata structure. With the objective of extending the financing-gap estimation to the widest possible set of countries, the analysis is conducted for all EU Member States, thereby expanding the Eurozone sample used in the Commission report. Any additional restrictions arising from data availability across survey waves are applied consistently across the different financing-gap variants, as defined in chapter 5 (all filters are described in detail in Section 3.1). After implementing all relevant re-

[2] Hereafter, all descriptive statistics and aggregate financing-gap measures reported are computed using these weights. It's important to note that the survey and the weights do not consider SMEs with zero employees.

strictions, the final sample consists of 89 150 observations covering the 27 EU countries. The distribution of firms by country, sector, and size is reported in Tables 1 and 2 below. [3]

Table 1: Distribution of observations by country

Country	**Freq.**	**%**	**Country**	**Freq.**	**%**	**Country**	**Freq.**	**%**
AT	3 362	3.77	DE	7 782	8.73	LT	1 275	1.43
BE	4 216	4.73	DK	895	1.00	LU	252	0.28
BG	1 678	1.88	EE	308	0.35	LV	699	0.78
CY	384	0.43	ES	9 494	10.65	MT	301	0.34
CZ	1 393	1.56	FI	2 481	2.78	NL	3 848	4.32
FR	11 323	12.70	GR	7 041	7.90	PL	3 871	4.34
HR	1 144	1.28	HU	1 827	2.05	PT	4 303	4.83
IE	2 467	2.77	IT	11 787	13.22	RO	2 336	2.62
SE	1 025	1.15	SI	762	0.85	SK	2 904	3.26

Table 2: Firm characteristics

(a) Sector distribution

Sector	Freq.	%
Industry	20 382	22.86
Construction	10 381	11.64
Trade	20 248	22.71
Services	29 475	33.06
NA	8 672	9.73

(b) Size distribution (No. of employees)

Size class	Freq.	%
$1 \leq$ size ≤ 9	32 586	36.55
$10 \leq$ size ≤ 49	25 162	28.22
$50 \leq$ size ≤ 249	22 738	25.50
size ≥ 250	8 672	9.73

[3] The total number of observations comprises both firms with observed loan applications and discouraged shadow firms, in accordance with the broadest sample definition adopted for Method 3 (see Chapter 5). Unless otherwise stated, all reported figures and graphs will follow the same sample construction.

2.2 Estimation Methods

This section describes how firm-level financing gaps are constructed from SAFE. Following the Commission approach, the financing gap is recovered by combining two components: i) a **financing-need factor** (capturing whether the firm has no, partial, or full unmet need), and ii) a **euro-valued loan-size component** (converting the categorical need into a euro amount). These two components are then multiplied as follow to derive firm-level financing gaps:

$$\text{Gap}_{it} \;=\; NeedFactor_{it} \cdot LoanSize_{it}, \tag{1}$$

where $NeedFactor_{it} \in [0, 1]$ captures the intensity of unmet financing need and $LoanSize_{it} \geq 0$ is the requested loan amount expressed in euros (proxied by the midpoint of the Q8A loan-size bracket). Intuitively, the $NeedFactor$ captures "how much of the request is unmet," while the $LoanSize$ provides the scale needed to express the gap in euros.

2.2.1 Financing-need factor (based on Q7A and Q7B)

The first step assigns each firm a factor between 0 and 1 that represents the intensity of unmet financing need based on Q7A and Q7B. I assume that a firm has no financing need if it has applied to a bank loan and received the full amount, or if it did not even apply to a bank loan because of sufficient internal funds (NeedFactor = 0). On the other end, I assume that a firm has a full financing need if its application for a bank loan was rejected or if the firm refused it because the offered conditions were unfavourable (NeedFactor = 1). In cases where a firm got "a limited part" or "most of" the bank loan they applied for, following Commission Report approach, I assume that the firm has a partial financing need of 80% or 20%, respectively (factor 0.8 or 0.2).

Table 3: Distribution of questions Q7A and Q7B on the application for bank loan financing

Survey question	Question's categories	No. of observations	Percentage share	Needfactor
Q7A Application for a bank loan	1 – Applied	48 162	54.02	–
	2 – Did not apply because of possible rejection	9 247	10.37	1
	3 – Did not apply because of sufficient internal funds	3 343	3.75	0
Q7B Outcome of the bank loan application	1 – Applied and got everything	33 349	37.40	0
	3 – Applied but refused because cost too high	779	0.87	1
	4 – Applied but was rejected	3 068	3.44	1
	5 – Applied and got most of it	3 166	3.55	0.2
	6 – Applied and got a limited part of it	2 603	2.92	0.8

2.2.2 Euro conversion via requested loan size (based on Q8A)

The second step converts the categorical need factor into a euro amount using the requested loan size. The survey records the requested loan size in the question Q8A as an ordinal variable with five bracket ranges. Each bracket is mapped to a midpoint (see Table 4). For the open-ended top bracket ("over EUR 1 million"), the approach assumes a fixed representative value (EUR 1.5 million), consistent with the approach used by **beck-2023**.

Table 4: Overview of question Q8A on the size of the last requested bank loan

Q8A categories	Midpoint	No. of observations	Percentage share
1 – up to EUR 25 001	12 500	4 160	4.67
2 – more than EUR 25 000 and up to EUR 100 001	62 500	9 106	10.21
4 – more than EUR 100 000 and up to EUR 250 001	175 000	10 229	11.47
5 – more than EUR 250 000 and up to EUR 1 million	625 000	9 834	11.03
6 – over EUR 1 million	1 500 000	1158	1.30
NAs	–	47 545	53.33

2.2.3 ”Shadow Firms” and discouraged borrowing (based on Q32)

As a key feature of this methodology, I include discouraged firms that have financing needs but refrain from applying due to expected rejection or market barriers. These ”shadow firms” are indentified by combining i) Q7A: firms indicating they did not apply because of possible rejection, and ii) Q32: firms reporting bank loans as not relevant for reasons consistent with market barriers (all Q32 answers excluding the responses 8, 9 – see Table 5).

Table 5: Non-relevance of Bank loans: reasons (Q32)

Variable name and question	**Categories**	**No. of observations**	**Percentage**
Q32 Non relevance reason for bank loans	1 – Insufficient collateral or guarantee	4 954	3.25
	2 – Interest rates or price too high	11 901	1.87
	3 – Reduced control over the enterprise	1 288	0.08
	4 – No bank loans are available	4 477	3.57
	5 – Other	51	0.59
	6 – Too much paperwork is involved	63	0.73
	8 – I do not need this type of financing	7 745	89.17
	9 – I do not know	66	0.76

For firms classified as shadow firms, I assume a full unmet financing need (NeedFactor = 1). Translating that assumption into a monetary financing gap requires a counterfactual requested loan amount; the method used to handle and impute this missing loan-size component is detailed in the next Chapter.

Chapter 3

Recovering Missing Data – A Machine Learning Approach

This chapter examines how missing information in the SAFE survey is handled when constructing euro-valued firm-level financing gaps. The focus is on imputing the loan-size component from question $Q8A$, which is necessary to translate reported financing needs into monetary amounts. Building on the conceptual framework introduced in Chapter 3, this chapter addresses the empirical challenges posed by missing and discontinued loan-size data and outlines the methodology used to overcome them.

Two related data issues arise with respect to question $Q8A$. First, even in survey waves where the question is included, item non-response is non-negligible. Missing values occur due to "don't know / no answer" responses, routing filters, and inconsistencies in survey completion, resulting in incomplete loan-size information within otherwise usable observations. Second, and more importantly, $Q8A$ is no longer collected in recent SAFE waves. After being introduced in wave 11 (H1 2014), the question was discontinued after wave 28 (H2 2022). From 2023 onwards, the SAFE survey therefore lacks direct information on loan sizes, even though all the remaining relevant questions required to identify financing needs and application outcomes continue to be asked. Without corrective action, this discontinuity would prevent the computa-

tion of euro-valued financing gaps for the most recent years and would effectively truncate the analysis at 2022. Extending financing-gap measurement beyond 2022 requires addressing the missing $Q8A$ information; this chapter therefore implements a machine-learning imputation of the missing loan-size component. Within this framework, two alternative imputation strategies are compared (Chapter 3.2.2: (i) imputing the categorical $Q8A$ loan-size responses and subsequently mapping them to euro midpoints, and (ii) directly imputing the loan-size midpoints without predicting the underlying $Q8A$ categories. While the distinction may appear minor at first glance, it leads to materially different model implementations and has non-trivial implications for how the imputation techniques are applied and evaluated.

3.1 Data Cleaning and Preparation

3.1.1 Sample construction and identification of bank-loan demand

This subsection describes the sample-construction steps performed prior to imputing the loan-size component, yielding 89 158 usable observations. [4] The SAFE microdata were cleaned along three dimensions – (i) firm demographics, (ii) bank-loan demand, and (iii) firm viability – so that the machine-learning model is trained on the population for which the loan-size question is conceptually meaningful.

A first demographic pruning step restricts the sample to EU countries and to observations starting from wave 11 (H1 2014), i.e. the first wave in which $Q8A$ was introduced on a consistent basis. From 2024Q1, quarterly waves are aggregated into semester equivalents (Q1-Q2 and Q3-Q4) and observations are deduplicated so that each firm contributes at most one record per half-year.

A second sample restriction is performed to effectively retain only firms that have positive bank-loan demand, which is formed by i) *observed demand* ($Q7A$, answers 1 and 2) and ii) *shadow*

[4] See appendix A.1.

demand or discouraged demand ($Q32$, excluding answers 8 and 9).[5]

3.1.2 Predictors selection

After defining the estimation sample, I construct the set of predictors used by the machine-learning imputation models. As a first step, I harmonise the variables stored as labelled field into factors to enable a consistent treatment as categorical variables. For each of these factor predictors, instead of dropping rows, missing values are explicitly encoded as a dedicated "*MISSING*" category, so as to exploit the non-responses as a potentially informative signal. Then, in a second step, I reduce the number of predictors from 622 recorded predictors to 186, intending to reduce noise and improve stability. First, all predictors with missingness over 90% are removed, then a data-driven screening rule is implemented: a predictor is retained if the missingness pattern of that variable is statistically related to the target.[6] This rule preserves variables whose non-response contains relevant information to predict loan size.

3.1.3 Target-leakage control

To ensure a valid out-of-sample evaluation and to prevent mechanical predictability, the imputation models exclude any variables that directly encode the target information. In model I, where $Q8A$ is the target, only its recoded version is dropped ($Q8A_rec$). On the other hand, for model II, where the target variable is $Q8A_midpoint$, both $Q8A$ and $Q8A_rec$ are dropped from the training set.

3.1.4 Q8A missingness and pre-imputation distribution

Before starting with the imputation part, $Q8A$ is cleaned by recoding "DK/NA" (category 9) to missing (NA), since it does not correspond to any bracket and cannot be mapped to a

[5]Note: firms reporting outcomes in $Q7B$ are, by construction, a subset of firms that report having applied in $Q7A$; therefore, $Q7B$ does not add a "demand filter" but only generates the financing-need factor.

[6]See appendix A.2.

midpoint. Therefore, missingness has three main sources. i)Item non-response within waves where $Q8A$ is asked (e.g., skipped or incomplete answers). ii) Structural missingness after the question is discontinued: Table 6 shows that missingness is fairly stable across waves 11 – 28, and then becomes 100% from wave 29 onward because $Q8A$ is no longer collected. iii) By construction, regarding the shadow firms, where, since these firms did not apply for a bank loan, the question $Q8A$ is not asked. For this kind of firm, $Q8A$ is therefore imputed with a counterfactual interpretation: the model predicts the loan-size bracket the firm would plausibly have applied for, had it applied, conditional on its observed characteristics and answers. The country/firm-characteristic tables do not indicate major irregularities, but some moderate and systematic heterogeneity, with a clear size gradient as shown in 8b (higher missingness among micro firms, lower among larger firms). Sector differences are present but comparatively mild, with industry showing a lower missing share than construction/trade/services (Table 8a).

Table 6: NAs distribution by wave

Wave	**Obs.**	**Percent**	**Wave**	**Obs.**	**Percent**	**Wave**	**Obs.**	**Percent**
11	2 882	50.7	19	2 280	48.4	27	2 269	53.6
12	2 022	47.7	20	1 572	43.6	28	1 520	49.2
13	2 802	51.6	21	2 276	50.7	29	4 177	100
14	1 863	45.2	22	1 434	40.8	30	2 147	100
15	2 485	49.0	23	2 106	39.9	32	2 761	100
16	1 735	44.0	24	1 406	40.4	34	1 452	100
17	2 383	49.4	25	1 949	49.7	36	2 096	100
18	1 658	43.6	26	1 428	46.1			

Table 7: NAs distribution by country

Country	Obs.	Percent	Country	Obs.	Percent	Country	Obs.	Percent
AT	1 780	52.9	DK	634	70.8	LU	144	57.1
BE	1 600	38.0	EE	228	74.0	LV	537	76.8
BG	1 227	73.1	ES	4 126	43.5	MT	170	56.5
CY	262	68.2	FI	1 047	42.2	NL	2 615	68.0
CZ	800	57.4	FR	4 459	39.4	PL	2 663	68.8
DE	4 326	55.6	GR	5 113	72.6	RO	1 839	78.7
HR	671	58.6	HU	1 424	77.9	SE	598	58.3
IE	1 357	55.0	IT	5 408	45.9	SI	403	52.9
LT	906	71.1	SK	1 934	66.6	PT	2 432	56.5

Table 8: NAs by firm characteristics

(a) Sector distribution

Sector	Freq.	%
Industry	9 850	48.3
Construction	6 261	60.3
Trade	11 680	57.7
Services	17 821	60.5
NA	3 091	35.6

(b) Size distribution (No. of employees)

Size class	Freq.	%
$1 \leq$ size ≤ 9	22 755	69.8
$10 \leq$ size ≤ 49	13 301	52.9
$50 \leq$ size ≤ 249	9 556	42.0
size ≥ 250	3 091	35.6

3.2 Catboost for Survey Imputation

In this chapter, I adopt a supervised machine learning approach to impute the loan-size information from SAFE ($Q8A$). The observed $Q8A$ responses provide the labelled outcome, while the remaining characteristics of the firm and the survey act as predictors. This setting features many categorical variables, potentially nonlinear relationships, and strong interactions across country, firm size, sector, and survey wave – patterns that are cumbersome to capture with parsimonious parametric specifications but can be learned flexibly in a predictive framework. The technique's choice is guided by the need to i) handle many categorical predictors without over-relying on *one-hot expansion*; ii) generalise stably in an out-of-time validation design; iii) handle rows with missing data without discarding them; iv) remain computationally feasible; v) support both categorical (model I) and continuous formulations of the target (model II).

To meet these requirements, I rely on CatBoost, a *gradient-boosting* algorithm over decision trees designed to handle categorical data while mitigating coding-induced leakage through *ordered target* encoding. In addition, CatBoost uses symmetric decision trees, which improves stability and computational efficiency.

3.2.1 Theoretical Framework of the CatBoost Algorithm

Following the original contribution of **prokhorenkova-2019**, who introduce CatBoost as an unbiased gradient boosting algorithm specifically designed for categorical features, the theoretical workflow of the method can be conceptually decomposed into three components.

(i) **Ordered target encoding of categorical predictors** The first step addresses the central challenge of categorical data by transforming them into a numerical format that can be fed to tree learners. CatBoost therefore converts each categorical feature into a numerical feature using ordered target statistics (this is performed separately for each categorical predictor $C^{(z)}$ in the feature set). Concretely, for a given predictor z, categorical level c, and firm i, the algorithm constructs a value that summarises how the target ($Q8A$) behaved historically among firms

with the same level c.

These raw category-level statistics are then smoothed toward the global target distribution, so that rare categories do not receive extreme values purely due to small-sample noise. A key feature is the sequential logic of how the statistics are computed. Instead of estimating them on the full training sample, which would cause target leakage, the encoding uses only informations from earlier observations in the dataset, ensuring that the numerical representation of a category is formed without using the observation's own target. The new encoded numerical value TE for every observation (row) i, predictor z (column), categorical level c is computed as:

$$\mathrm{TE}_i(c) = \frac{\sum_{j \in \mathcal{P}(i)} \mathbf{1}\{C_j = c\}\, Y_j \;+\; \alpha\, \mu}{\sum_{j \in \mathcal{P}(i)} \mathbf{1}\{C_j = c\} \;+\; \alpha}, \qquad \mu = \mathbb{E}[Y]. \tag{2}$$

where C denotes a specific categorical predictor (e.g. sector) with support $\mathcal{C} = \{c_1, \ldots, c_{|\mathcal{C}|}\}$, i.e. the set of all possible sector categories. [7] For each observation i, $C_i \in \mathcal{C}$ is the realized category level of that observation; writing $C_i = c$ means that firm i belongs to category level c, for what concerns the predictor C. The ordered target-encoding feature $\mathrm{TE}_i(c)$ is computed for observation i by aggregating information from other observations indexed by j, restricted to those that: i) precede i in the ordered permutation of the training sample, and ii) share the same category level c, i.e. $C_j = c$. Formally, $\mathcal{P}(i)$ denotes the set of indices j that come before i in the permutation; thus $\sum_{j \in \mathcal{P}(i)} \mathbf{1}\{C_j = c\}$ counts how many prior observations share the same category, while $\sum_{j \in \mathcal{P}(i)} \mathbf{1}\{C_j = c\}\, Y_j$ sums the corresponding transformed targets. The scalar $\alpha > 0$ controls smoothing (shrinkage) toward the prior μ, which is the global mean when Y is numerical ($\mu = \mathbb{E}[Y]$).

[7] The same construction applies to each categorical predictor $C^{(z)}$, $z = 1, \ldots, Z$; for notational simplicity the index z is omitted.

$$\mathrm{TE}_i^{(k)}(c) = \frac{\sum_{j \in \mathcal{P}(i)} \mathbf{1}\{C_j = c\}\, \mathbf{1}\{Y_j = k\} \;+\; \alpha\, \pi_k}{\sum_{j \in \mathcal{P}(i)} \mathbf{1}\{C_j = c\} \;+\; \alpha}, \qquad k = 1, \ldots, K, \;\; \pi_k = \Pr(Y = k). \tag{3}$$

When the target is categorical with K levels, $Y \in \{1, \ldots, K\}$, the encoding is computed *class-by-class*. In particular, the index $k \in \{1, \ldots, K\}$ denotes one specific target class (e.g. one$Q8A$bin). Y_j in the numerator of the regression case is replaced by the class indicator $\mathbf{1}\{Y_j = k\}$, so that $\sum_{j \in \mathcal{P}(i)} \mathbf{1}\{C_j = c\}\mathbf{1}\{Y_j = k\}$ counts how many prior observations with $C_j = c$ belong to class k. Likewise, the prior term μ is replaced by the global class prior $\pi_k = \Pr(Y = k)$. As a result, for each observation i and category level c, the procedure produces a vector of encoded features $\big(\mathrm{TE}_i^{(1)}(c), \ldots, \mathrm{TE}_i^{(K)}(c)\big)$, where each $\mathrm{TE}_i^{(k)}(c)$ can be interpreted as a smoothed estimate of the conditional class frequency $\Pr(Y = k \mid C = c)$, computed using only the preceding observations $j \in \mathcal{P}(i)$ to mitigate target leakage.

In simple terms, target encoding replaces each category with a single number: the average value of the target among past observations in that category. When there are few observations, this estimate is stabilised by shrinkage towards the global mean.

(ii) **Construction of the boosted tree ensemble**. The second (learning) step trains the model on the subset of rows for which the target variable is observed (i.e. non-missing). CatBoost is an implementation of *Gradient Boosting*, where the base learners are *binary-split decision trees*.[8] Each tree is built by recursively partitioning the feature space into disjoint regions using threshold rules on subsequently different predictors. After categorical variables have been mapped into numerical features through ordered target statistics, splits take the form of numerical inequalities (e.g. $X_i^{(z)} \leq \tau$, with $z \in \{1, \ldots, Z)$.[9]

[8] A decision tree is a rule-based predictor that iteratively splits the training sample into increasingly homogeneous groups using simple *if – then* conditions on the covariates. In this case, the trees are symmetric and always split into two complementary regions (binary).

[9] Here $X_i^{(z)}$ denotes the value of the m-th predictor (column) for observation i (row) after TE, and Z is the total number of predictors in the training set.

Starting from the root node, the algorithm considers candidate splits of the form $X_i^{(z)} \leq \tau$ across predictors z and thresholds τ, and selects the split that delivers the largest improvement in the objective, i.e. the largest reduction in the loss function used by the boosting procedure. Conditional on the learned partition, the algorithm assigns a constant prediction to each leaf (γ_ℓ). Let $X_i \in \mathbb{R}^Z$ denote the feature vector for firm i. A fitted tree $h(\cdot)$ can be written as

$$h(X_i) = \sum_{\ell=1}^{L} \gamma_\ell \, \mathbf{1}\{X_i \in R_\ell\},$$

where γ_ℓ is the *leaf value* (leaf output) for region R_ℓ. These leaf values are estimated from the observations that fall into each leaf to minimize the loss given the current boosting state: in regression (midpoint target) $\gamma_\ell \in \mathbb{R}$ is a scalar update (interpretable in euros), whereas in the multi-class specification ($Q8A$ bins) $\gamma_\ell \in \mathbb{R}^K$ is a vector of class scores that is subsequently mapped into predicted class probabilities. [10]

In gradient boosting, the model is not based on a single tree, but on an *ensemble* of many trees trained sequentially. The first tree provides an initial approximation of Y; then, each additional tree is fitted to explain what the previous trees still fail to capture – i.e., the remaining prediction error. As the boosting iterations proceed, different trees can rely on different predictors and split rules, so that the ensemble gradually incorporates multiple sources of information and captures nonlinearities and interactions without having to specify them explicitly. At the end of training, the contributions of all trees are combined to obtain the final prediction for each firm: intuitively, the ensemble aggregates the successive "corrections" learned by the individual trees. When Y is numerical (midpoint specification), the output is a single continuous prediction $\widehat{Y}_i$. When Y is categorical ($Q8A$-bin specification), the output is a vector of predicted class probabilities $\widehat{p}_{ik} = \Pr(Y_i = k \mid X_i)$, which is then translated into an imputed category by drawing a single

[10] γ_ℓ is the "best constant" update for observations assigned to leaf ℓ; when added to the current model prediction, it is chosen to minimise the objective loss, and the updated predictions then become the new baseline from which the next boosting iteration learns further corrections. The implemented loss functions vary: `loss_function = "MultiClass"` when the target is categorical ($Q8A$ loan-size bins) and `loss_function = "RMSE"` when the target is continuous (the euro-valued $q8a_midpoint$).

$Q8A$ bin from the predicted probability vector $\widehat{p}_{ik}$ for each firm, rather than deterministically selecting the most likely.

(iii) **Predictions of previously unseen observations**. Once the model is trained on the subsample with observed targets, it is applied to all firms for which $Q8A$ is missing. Given the encoded predictor vector X_i, the fitted ensemble returns either a continuous prediction $\widehat{Y}_i$ in the midpoint specification or a probability vector $\widehat{p}_{ik}$ over $Q8A$ bins in the multi-class specification. These predictions are then used to fill in missing values and construct a completed $Q8A$ series that can be consistently mapped into euro-valued loan sizes for the financing-gap computation.

3.2.2 Model specifications

The two imputation specifications used to reconstruct the requested loan-size component from SAFE question $Q8A$ share the same modelling pipeline -sample construction, predictor harmonisation, and estimation via gradient-boosted trees- and differ only in the definition of the prediction target. Throughout, estimation is implemented with the `CatBoost` package, which is well suited to the SAFE setting given the prevalence of categorical covariates and nonlinear interactions. Both methods are trained on the subset of observations where the target variable is observed, so that the algorithm can learn the empirical relationship between firm characteristics and the requested loan-size information. The training data retain the high-dimensional categorical structure of the SAFE survey; categorical predictors are handled directly within `CatBoost`, avoiding extensive one-hot encoding. The objective of both specifications is to produce a completed loan-size series that can be plugged into the euro-valued financing-gap formulas defined in Chapter 2, ensuring comparability with the survey's original bracket design. Finally, both specifications produce a completed loan-size series together with a midpoint mapping consistent with Chapter 2. Conceptually, they only differ in whether the model learns the brackets directly (Model I) or learns their midpoint representation and then maps predictions back to the admissible midpoint set (Model II).

Model I: categorical bracket imputation. Model I treats $Q8A$ as a *five-class categorical outcome*, corresponding to the SAFE loan-size brackets used for euro conversion. Estimation uses the multi-class classification loss `MultiClass`.[11] For each firm i, the model outputs a probability vector $\widehat{\mathbf{p}}_i = (\hat{p}_{i1}, \ldots, \hat{p}_{i5})$ over the five brackets, rather than a single deterministic label. The probabilistic output provides the basis for imputation: missing $Q8A$ values are filled via a stochastic draw from $\widehat{\mathbf{p}}_i$, rather than by assigning $\arg\max_k \hat{p}_{ik}$. This stochastic rule is preferred because deterministic argmax assignment tends to compress heterogeneity (over-allocating observations to the modal bracket and attenuating tail categories), whereas probability draws better preserve cross-sectional dispersion, which is crucial when the imputed series is converted into euro-valued financing gaps. Reproducibility is ensured by fixing the random seed used for the draws.

Model II: midpoint-target imputation. Model II follows the same overall structure but shifts the target from brackets to *euro midpoints*. Concretely, $Q8A$ is first mapped into the corresponding midpoint value m_k, and the model is trained to predict this midpoint directly as a continuous outcome. This specification therefore replaces the categorical objective with a regression objective (implemented in `CatBoost` using `RMSE`). To preserve comparability with the discrete SAFE design, predicted midpoint values are subsequently constrained to the set of admissible midpoints (i.e. predictions are "snapped" back to the five midpoint levels). This ensures that Model II yields euro values that remain fully consistent with the original bracket structure, while allowing the model to exploit the ordering and distance between brackets during training.

Class imbalance and weighting (both models). Loan-size categories are not equally frequent, and the upper brackets are relatively rare but economically important because they carry disproportionate weight in euro conversion. Both specifications therefore apply class

[11] See appendix A.4.

weights proportional to the inverse of class frequencies scaled by median class frequency, $w_k = \mathrm{median}(n_1, \ldots, n_K) \,/\, n_k$. This increases the penalty for errors associated with rare brackets, reducing the tendency to over-predict the modal category and improving performance in the tails of the implied euro distribution.

Hyperparameters and regularisation (both models). Both methods adopt a deliberately conservative tuning strategy. Rather than extensive optimisation, a small grid search is performed over a limited set of hyperparameters – tree depth, learning rate, and L_2 regularisation (`l2_leaf_reg`) – with early stopping based on an overfitting detector (`od_wait`).[12] This approach reflects two design principles: (i) *stability over marginal gains* in a survey setting with evolving composition, and (ii) *regularised flexibility*, where depth and L_2 govern model capacity while the learning rate controls how gradually the boosted ensemble updates. After selecting the configuration, the final model is re-estimated on all observations with an observed target to maximise information before imputing missing values.

[12] See appendix A.3.

Chapter 4

Model Validation and Selection

This chapter compares the two CatBoost-based imputation specifications introduced in Section 3.2.2 for reconstructing the requested loan-size component from SAFE question $Q8A$. The goal is twofold: (i) assess out-of-sample predictive performance under a time-aware evaluation design, and (ii) verify that the imputed series is *credible* in a survey sense, i.e., it does not distort the empirical distribution of requested loan-size brackets in a way that would mechanically bias downstream financing-gap aggregates.

Recall the two competing models:

- **Model I (categorical bracket imputation)**. $Q8A$ is modelled as a five-class categorical outcome. The fitted model returns a probability vector $\hat{\mathbf{p}}_i = (\hat{p}_{i1}, \ldots, \hat{p}_{iK})$ over the $K = 5$ brackets and imputes missing $Q8A$ via a stochastic draw from $\hat{\mathbf{p}}_i$ (rather than a deterministic argmax rule).
- **Model II (midpoint-target imputation)**. $Q8A$ is first mapped into the corresponding midpoint value m_k, and the model is trained to predict this midpoint directly as a continuous target under an RMSE loss.

A central issue for comparison is that the two models naturally live in different output spaces (probabilities over bins vs. continuous euros). The evaluation is therefore organised around three complementary spaces, ranked in order of importance for model selection:

1. **Bin space**. Performance in predicting the original Q8A brackets, accounting for ordinality. This includes both *ordinal accuracy* metrics (e.g. exact accuracy, within-1 accuracy, quadratic weighted kappa), *distributional credibility*, assessed via Total Variation Distance (TVD) between predicted and observed bracket shares, and *tail-risk metrics* that focus on relevant tail mistakes.

2. **Midpoint (euro) space** . Performance in predicting the euro-valued midpoint associated with each bracket. These metrics assess how ordinal differences translate into differences in predicted loan amounts.

3. **Gap space**. Differences in the resulting firm-level and aggregated financing-gap estimates implied by each imputation model. These metrics evaluate whether discrepancies in imputation meaningfully affect downstream economic quantities.

Model selection is based on an explicit analytical decision rule described in Section 4.5, which combines performance in bin space and euro space. Gap-space diagnostics are not used for ranking the models, but are reported as sanity checks to confirm that the chosen specification does not generate economically implausible distortions in the resulting financing-gap measures.

4.1 Harmonised evaluation design

To ensure that the comparison isolates the impact of the target definition (brackets vs. midpoints), both models share the same core pipeline:

- *Same sample restrictions and harmonisation.* Both specifications are estimated on the same cleaned SAFE sample and use the same categorical handling (explicit encoding of missing factor levels).
- *Same leakage control.* Predictors exclude $Q8A$ itself and any variables mechanically derived from it (e.g. recoded versions or midpoint-derived fields).

- *Same time-aware holdout logic.* Out-of-sample performance is evaluated using a wave-based split: the last $\approx 20\%$ of waves are held out as a test period, and the model is trained on earlier waves only. This is important in a survey environment with evolving composition, because random i.i.d. splits can overstate performance when there is temporal dependence. [13]
- *Class-imbalance correction.* Both models incorporate weights that upweight rare brackets, which reduces over-allocation to the modal bracket and improves tail behaviour.

All metrics reported below are computed on the *holdout waves* using only observations with observed $Q8A$ (or observed midpoints derived from $Q8A$). This ensures that model ranking is based on genuine predictive ability, not on internal consistency of the imputation step.

4.2 Cross-Specification Output Harmonisation

A direct comparison between the two specifications is not immediate because they generate predictions in different outcome spaces: Model I is formulated in terms of discrete loan-size *brackets*. while, Model II operates in a continuous *euro* scale. Therefore, the evaluation is conducted after an explicit harmonisation step: for euro-space diagnostics (4.2.1), Model I is mapped into a continuous prediction through the probability-weighted expected midpoint; conversely, for bin-allocation diagnostics (4.2.2), Model II is discretised by *snapping*[14] its continuous prediction to the nearest admissible SAFE midpoint (and hence to a bracket).

Let the five SAFE loan-size brackets be indexed in the fixed ordinal order $\mathcal{K} = \{1, 2, 5, 6, 4\}$, where the labels match the SAFE coding but are arranged in increasing loan-size order (so that the largest bracket is coded as 4 in SAFE, but placed last in the ordinal ordering).

[13] The wave-based split is used exclusively to assess out-of-time predictive performance (and to select hyperparameters via a one-shot validation). After the specification is fixed, the imputation model is refit on the full sample of observations with non-missing targets, and this refitted model is used to impute missing loan-size information. As a consequence, it is expected that the actual final model behaves better than the tested one, giving that it trains on all available waves.

[14] "Snapping" means mapping a continuous prediction to the closest admissible SAFE midpoint. Formally, given the finite set of midpoints $\{m_k\}$, the snapped value is the midpoint whose distance $|\hat{m} - m_k|$ is smallest. In the unlikely possibility that a prediction lies exactly halfway between two midpoints, it is snapped to the lower point.

4.2.1 Euro-space alignment (Model I)

When computing euro-space evaluation metrics, Model I probabilities are mapped into a euro-valued prediction via the probability-weighted expected midpoint:

$$\hat{m}_i^{(I)} = \sum_{k \in \mathcal{K}} \hat{p}_{ik}\, m_k \tag{4}$$

Because SAFE midpoints lie on a discrete support, for a fair evaluation, it is necessary to compute a *snapped* midpoint:

$$\tilde{m}_i^{(I)} = \arg \min_{m \in \{m_k : k \in \mathcal{K}\}} \left| \hat{m}_i^{(I)} - m \right| \tag{5}$$

Hereafter, all reported euro-space and log-euro-space metrics for Model I are computed using $\tilde{m}_i^{(I)}$ as in equation (5). The final imputation step samples stochastically from $\hat{\mathbf{p}}_i$ to preserve the implied distribution. For holdout evaluation, however, stochastic draws would introduce Monte Carlo noise into evaluation metrics, making results depend on the random seed rather than purely on predictive skill. The *argmax* rule, therefore, yields fully reproducible metrics that better isolate the model's learned signal.

4.2.2 Bin-space alignment (Model II)

Model II produces a continuous midpoint prediction $\hat{m}_i^{(II)}$. The script then: (i) *clamps* [15] predictions to the admissible midpoint range, and (ii) *snap* to the nearest SAFE midpoint:

$$\tilde{m}_i^{(II)} = \arg \min_{m \in \{m_k : k \in \mathcal{K}\}} \left| \text{clamp}(\hat{m}_i^{(II)}) - m \right|, \tag{6}$$

[15] The clamp operator projects a real value onto a bounded interval: for $a \leq b$, $\mathrm{clamp}(x; a, b) = \min\{\max\{x, a\}, b\}$, i.e. it equals a if $x < a$, equals x if $a \leq x \leq b$, and equals b if $x > b$. In our case $a = \min_k m_k$ and $b = \max_k m_k$.

where clamp$(\cdot)$ truncates values to $[\min_k m_k, \max_k m_k]$. The implied predicted bracket $\tilde{k}_i^{(II)}$ is the bracket corresponding to $\tilde{m}_i^{(II)}$.

Implication for comparability. After these steps, both models yield: i) a discrete predicted bracket on the same ordinal support, and ii) a snapped midpoint in euros on the same five-point SAFE midpoint support, that is exactly the object used in the financing-gap euro conversion to construct the financing-gap.

4.3 Evaluation metrics

In comparing the two models, the evaluation metrics are organised into three spaces, reflecting the fact that the models produce outputs in different formats (bins vs. euros) and that the ultimate object of interest is the financing gap.

4.3.1 Bin-space metrics

Let $k_i \in \{1, \ldots, K\}$ denote the *ordered bracket index* of firm i's observed Q8A response. To construct k_i, the K SAFE loan-size brackets are first sorted in increasing order of their midpoints (from the smallest requested amount to the largest) and then re-labelled with consecutive integers $\{1, \ldots, K\}$. We set $k_i = k$ if firm i reports the k-th bracket in this ordered list, so that larger values of k_i correspond to larger requested loan sizes. Let $\hat{k}_i$ be the predicted ordered index (Model I: $\hat{k}_i = \arg\max_k \hat{p}_{ik}$; Model II: $\hat{k}_i$ is the bracket implied by the *snapped midpoint*).[16]

Ordinal performance is then assessed through:

- *Exact accuracy*: $\Pr(\hat{k}_i = k_i)$

[16]Bin-space predictions rely on the modal class $\hat{k}_i = \arg\max_k \hat{p}_{ik}$, which is consistent with the multi-class log-loss objective used in training. Assigning bins via the probability-weighted expected midpoint and subsequent snapping would effectively impose a regression-based rule, misaligned with the classification loss, and would render Model I artificially similar to the continuous specification of Model II.

- *Within-1 accuracy*: $\Pr(|\hat{k}_i - k_i| \leq 1)$
- *Quadratic weighted kappa (QWK)*: an agreement measure for ordered categories that compares the predicted and true brackets while assigning larger penalties to larger ordinal errors. Misclassifying a firm by two or more brackets is therefore treated as more severe than a one-bracket mistake; QWK equals 1 under perfect agreement and is close to 0 under chance-level agreement.
- *Macro F1*: the unweighted average of the class-specific F1-scores across the K brackets. For a given bracket k, the F1-score is the *harmonic mean* of *precision* and *recall*, $F1_k = 2\,P_k R_k/(P_k + R_k)$. *Precision* P_k measures how reliable predictions of bracket k are, $P_k = \mathrm{TP}_k/(\mathrm{TP}_k + \mathrm{FP}_k)$,and therefore mainly penalises *false positives* (firms predicted as k but truly in another bracket). *Recall* R_k measures how many true bracket-k firms are correctly recovered, $R_k = \mathrm{TP}_k/(\mathrm{TP}_k + \mathrm{FN}_k)$, and therefore mainly penalises *false negatives* (firms truly in k but predicted elsewhere). Macro F1 averages $F1_k$ over brackets, so each bracket receives equal weight regardless of its frequency, making the measure informative under class imbalance. The score ranges from 0 to 1, with higher values indicating better overall classification performance across all brackets.

Distributional credibility Accuracy alone is insufficient for survey imputation: a model may achieve acceptable individual-level accuracy while distorting the bracket distribution in ways that bias aggregates. For any group g (overall, by wave, by country), let s_{gk} be the empirical share in bracket k and $\hat{s}_{gk}$ the predicted share. Total Variation Distance (TVD) is:

$$\text{TVD}_g = \frac{1}{2} \sum_{k \in \mathcal{K}} |\hat{s}_{gk} - s_{gk}| \,. \tag{7}$$

Finally, it is reported the *mean shift* in the ordinal index within each group (predicted minus true mean), which is a compact indicator of systematic upward/downward bracket bias.

Tail-risk Since financing-gap aggregates are sensitive to tail behaviour, it is necessary to compare the models also under these risk metrics:

- *Big mistake rate:* $\Pr(|\Delta\text{bin}| \geq 2)$. The share of observations for which the predicted bracket is at least two ordinal steps away from the true bracket.
- *Top-bin underprediction rate:* $\Pr(\hat{k}_i < K \mid k_i = K)$. Among firms truly in the highest bracket ($k_i = K$), the probability of predicting any lower bracket.
- *Catastrophic flip rate:* the share of observations where the prediction flips an extreme into the opposite extreme, i.e. predicting the bottom bracket when the true bracket is the top one or vice versa ($k_i = 1, \hat{k}_i = K$ or $k_i = K, \hat{k}_i = 1$).

4.3.2 Euro-space and log-euro-space metrics

Let $m(y_i)$ be the true midpoint implied by observed $Q8A$, $\tilde{m}_i$ the *snapped* predicted midpoint (as in Chapter 4.2), and $\bar{m}$ the *average* real midpoint across observations. Being the distribution right-skewed, the following metrics are also computed in the log-euro-space for which m_i and $\tilde{m}_i$ are respectively replaced by $\log(1 + m_i)$ and $\log(1 + \tilde{m}_i)$.

- *RMSE (€)*, *MAE (€)*, *Median AE (€)*

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\big(\tilde{m}_i - m_i\big)^2}, \quad \text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\big|\tilde{m}_i - m_i\big|, \quad \text{Median AE} = \text{median}\big(\big|\tilde{m}_i - m_i\big|\big)$$

- R^2, *Spearman correlation* (rank fit): Spearman correlation is computed between the ranks of predicted and observed midpoints and measures how well the model preserves the relative ordering of firms, independently of exact euro magnitudes.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(\tilde{m}_i - m_i)^2}{\sum_{i=1}^{n}(m_i - \bar{m})^2}, \quad \rho_S = \text{corr} \quad \big(\text{rank}(\tilde{m}_i),\, \text{rank}(m_i)\big)$$

- *Bias (mean/median)*, where the sign indicates bias' direction

$$\text{Bias}_{\text{mean}} = \frac{1}{n}\sum_{i=1}^{n}(\tilde{m}_i - m_i), \qquad \text{Bias}_{\text{median}} = \text{median}(\tilde{m}_i - m_i)$$

4.3.3 Gap-space metrics

Gap-space diagnostics evaluate the extent to which each imputation model reproduces the true euro-valued financing gap in the holdout period (i.e., the last waves reserved for out-of-time testing). For every firm i in the holdout sample, we compute the observed financing gap G_i using the actually reported $Q8A$ bracket (mapped into euros as in the financing-gap construction), and the model-implied gap $\hat{G}_i$ by replacing $Q8A$ with the model's imputed value. Metrics then compare $\hat{G}_i$ against G_i. The following holdout metrics are reported:

- *RMSE (€)*, *MAE (€)*, and *Median AE (€)*

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{G}_i - G_i)^2}, \quad \text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\hat{G}_i - G_i|, \quad \text{Median AE} = \text{median}\Big(|\hat{G}_i - G_i|\Big)$$

  with the *median AE* that is robust to the heavy right tail typical of euro gaps.

- R^2, *Spearman correlation* (rank fit): Spearman correlation is computed between the ranks of predicted and observed midpoints and measures how well the model preserves the relative ordering of firms financing gap, independently of exact euro magnitudes.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(\hat{G}_i - G_i)^2}{\sum_{i=1}^{n}(G_i - \bar{G})^2}, \quad \rho_S = \text{corr} \quad \big(\text{rank}(\hat{G}_i),\, \text{rank}(G_i)\big)$$

- *Bias (mean and median, €)*:

$$\text{Bias}_{\text{mean}} = \frac{1}{n}\sum_{i=1}^{n}(\hat{G}_i - G_i), \qquad \text{Bias}_{\text{median}} = \text{median}\Big(\hat{G}_i - G_i\Big),$$

  where the sign indicates systematic over- or underestimation.

- *Relative error (mean and median)*:

$$\mathrm{RelErr}_{\text{mean}} = \frac{1}{n}\sum_{i=1}^{n}\frac{|\hat{G}_i - G_i|}{|G_i| + \varepsilon}, \qquad \mathrm{RelErr}_{\text{median}} = \mathrm{median}\left(\frac{|\hat{G}_i - G_i|}{|G_i| + \varepsilon}\right),$$

where $\varepsilon > 0$ is a small constant introduced to avoid division by zero when $G_i = 0$.

4.4 Holdout comparison results

4.4.1 Accuracy, distributional, and tails comparison (bin-space)

Table 9: Out-of-time performance on ordered brackets (bin-space)

Metrics	**Model I**	**Model II**
Accuracy metrics		
Exact accuracy	0.5207	0.4793
Within-1 accuracy	0.8899	0.9069
Quadratic weighted kappa	0.7326	0.6917
Macro F1	0.4876	0.4208
Distributional credibility: bracket-share comparisons		
Overall TVD (bins)	0.0585	0.2290
Avg. TVD by wave	0.0686	0.2309
Avg. TVD by country	0.1478	0.2492
Avg. TVD by sector	0.1240	0.2380
Avg. TVD by size	0.1087	0.1854
Mean Shift	0.0246	0.1857
Tail-risk diagnostics		
Big mistake rate ($\lvert\Delta\text{bin}\rvert \geq 2$)	0.1101	0.0931
Top-bin underprediction rate	0.2564	0.3889
Catastrophic flip rate (bottom↔top)	0.0031	0.0008

Notes: All metrics are computed on the wave-holdout test period (the last 20% of waves).

Accuracy performance is overall better in Model I. It achieves higher *exact bracket accuracy* (0.5207 vs. 0.4793) and higher ordinal agreement as measured by *QWK* (0.7326 vs. 0.6917). *Macro F1* is also higher for Model I (0.4876 vs. 0.4208), consistent with better balance across

brackets under class imbalance. Model II has a modest advantage in *within-1 accuracy* (0.9069 vs. 0.8899), indicating that its errors are slightly more concentrated in neighbouring bins even though strict exact recovery is weaker.

Distributional credibility strongly favours Model I. The overall TVD is 0.0585 for Model I versus 0.2290 for Model II. The same pattern holds for average TVD by wave (0.0686 vs. 0.2309), by country (0.1478 vs. 0.2492), by sector (0.1240 vs. 0.2380), and by size (0.1087 vs. 0.1854). In other words, the regression-plus-snap approach substantially reshapes the predicted bracket composition relative to the observed composition in the holdout period, which is problematic for a financing-gap application where aggregation over groups is central.

Tail-risk diagnostics report mixed results. Model II reduces the frequency of very large ordinal errors (big mistake rate 0.0931 vs. 0.1101) and yields a lower catastrophic flip rate (0.0008 vs. 0.0031). However, Model II underpredicts the top bracket far more often (0.3976 vs. 0.2490), which is consequential because the top bracket anchors the right tail of the euro distribution and can materially affect aggregate financing gaps.

4.4.2 Midpoints comparison (euro-space)

Euro-space and log-Euro-space performance favour Model I even if slightly. The two specifications perform very similarly in RMSE, but Model I is consistently (if slightly) better across several diagnostics: RMSE is marginally lower (422 352.4 vs. 420 617.1), MAE is lower (242 606.3 vs. 243 902.7), and Spearman is higher (0.7414 vs. 0.7401). Median AE is identical (50 000), which is unsurprising given the discrete midpoint support. On the log scale – which downweights the mechanical dominance of the top bracket – Model I again performs better (RMSE 1.0701 vs. 1.0813; R^2 0.4939 vs. 0.4833), reinforcing the interpretation that the Multi-Class formulation preserves proportional fit more reliably across the distribution.

Table 10: Out-of-time performance on euro-valued midpoints (euro-space)

Metrics	**Model I**	**Model II**
€-space (snapped midpoints)		
RMSE (€)	422 353.4	420 617.1
MAE (€)	242 606.3	243 902.7
Median AE (€)	50 000	50 000
Spearman	0.7414	0.7401
R^2	0.4908	0.4950
Bias mean (€)	-18 847.4	-25 502.7
Bias median (€)	0.00	0.00
log1p(€)-space (snapped midpoints)		
RMSE (log1p)	1.0701	1.0813
MAE (log1p)	0.07014	0.7120
Median AE (log1p)	0.8755	0.8755
Spearman (log1p)	0.7414	0.7401
R^2 (log1p)	0.4939	0.4833
Bias mean (log1p)	0.2746	0.2487
Bias median (log1p)	0.00	0.00

Notes: All metrics are computed on the wave-holdout test period (the last 20% of waves).

4.4.3 Gap comparison (gap-euro-space)

Table 11: Out-of-time performance on final loan-gap (euro-gap space)

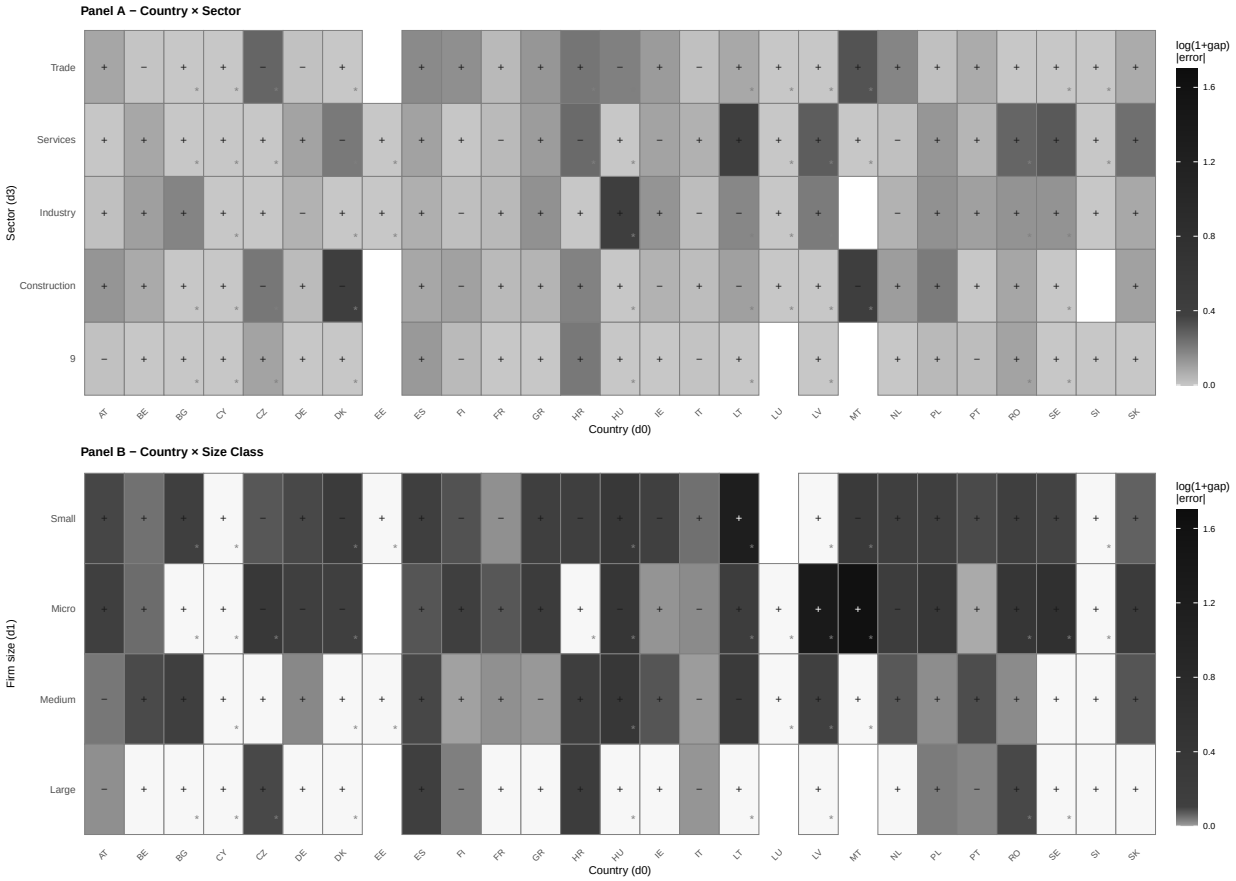
Metrics	**Model I**	**Model II**
RMSE (€)	149 147.8	152 072.9
MAE (€)	33 907.5	35 190.3
Spearman	0.9953	0.9951
R2	0.6283	0.6135
Real Error Mean	1.4752	1.5076
Real Error Median	0.00	0.00
Bias mean	-14 814.9	-13 934.8
Bias median	0.00	0.00

Notes: All metrics are computed on the wave-holdout test period (the last $\approx$ 20% of waves). The euro-gapp measures are computed considering also shadow firms, corresponding to *Methods 2 and 4* as described in Chapter 5.

Model I performs well in reproducing the observed financing gap out-of-time, and is better than Model II. Error magnitudes are moderate (RMSE €149 147.8; MAE €33 907.5), and the much larger RMSE than MAE suggests that performance is driven by a few big mistakes rather than

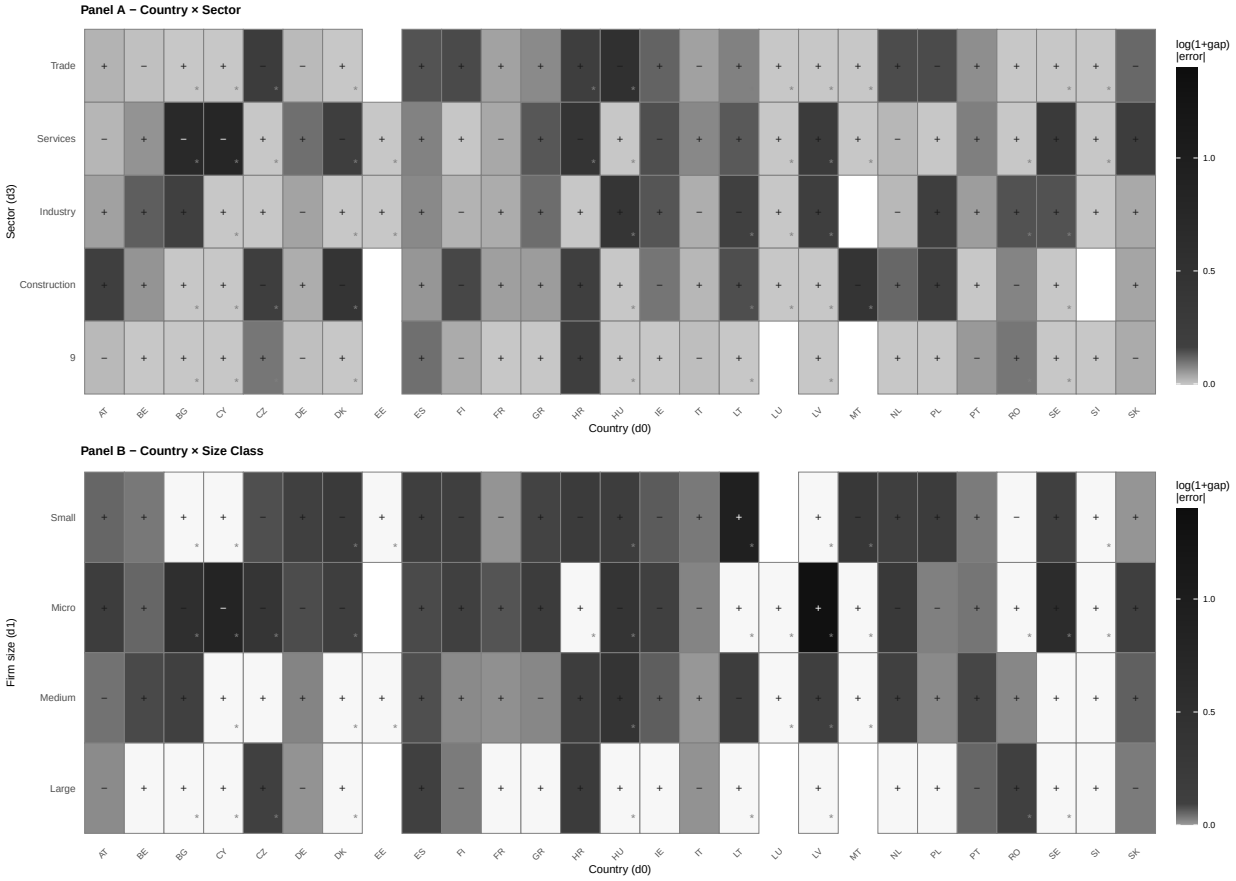
widespread error. The median absolute error is zero, meaning that at least half of the holdout gaps are predicted exactly; equivalently, this also explains the very high Spearman correlation (0.9953). Bias is limited for both models but slighty better for Model II (-13 934.8 vs. -14 814.93), indicating no systematic shift. Relative errors should be read with caution: the median relative error is zero for the same reason (many exact predictions), while the mean relative errors are high (1.4752 and 1.5076) because a small number of large errors-often occurring when the true gap is small or when a tail observation is missed – inflate percentage-based measures.

Figure 1: Model I – Mean log(1+gap) error by country, sector and size class (holdout waves)



Note: Cells report the mean error in $\log(1 + \text{gap})$ (predicted minus observed) computed over the holdout period (last $\approx 20\%$ of waves). Darker shading indicates larger absolute errors, while the sign of the error is shown by the $(+/-)$ marker. Asterisks denote cells with fewer than 10 holdout observations and should be interpreted with caution.

Figure 2: Model II – Mean log(1+gap) error by country, sector and size class (holdout waves)



Note: Cells report the mean error in $\log(1 + \text{gap})$ (predicted minus observed) computed over the holdout period (last $\approx 20\%$ of waves). Darker shading indicates larger absolute errors, while the sign of the error is shown by the $(+/-)$ marker. Asterisks denote cells with fewer than 10 holdout observations and should be interpreted with caution.

Cell-level gap errors (heatmap evidence). The heatmaps of log(1+gap) errors (predicted minus observed) confirm and refine the aggregate diagnostics. In both specifications, most country-sector and country-size cells display errors close to zero, suggesting the absence of a widespread systematic bias. Mis-predictions are concentrated in a limited number of small cells, often those with very low sample size, where volatility is mechanically higher. Large positive errors (overestimation) tend to appear in some micro-firm segments and smaller economies, while negative errors (underestimation) are more frequent in selected core-country × sector combinations. Relative to Model I, Model II produces a slightly smoother pattern in intermediate cells, but it does not eliminate tail-driven distortions: the same structurally fragile cells remain responsible for the largest deviations, with differences mainly in magnitude rather than direction. Overall, Model I appears more stable in preserving the cross-cell structure of the financing gap.

4.5 Rule-Based Model Selection

This chapter has compared the two CatBoost-based specifications along three evaluation spaces – bin space, midpoint (euro) space, and gap (euro) space – under a strictly out-of-time design. The objective is not merely predictive accuracy at the individual level, but the construction of a financing-gap series that is distributionally credible and economically coherent when aggregated across waves, countries, sectors, and firm characteristics. To make the implied trade-off explicit, model selection is implemented via a constrained decision rule: let $\mathcal{M} = \{\text{Model I}, \text{Model II}\}$ and, for each $m \in \mathcal{M}$, compute holdout metrics TVD_m (bin-space distributional distance), RMSE_m and MAE_m (midpoint euro-space errors), QWK_m and $\mathrm{W1}_m$ (ordinal agreement and within-one-bin accuracy), and TBU_m (top-bin underprediction rate). Define the "best" performance on each guardrail metric as $\mathrm{RMSE}^\star = \min_{k\in\mathcal{M}} \mathrm{RMSE}_k, \quad \mathrm{MAE}^\star = \min_{k\in\mathcal{M}} \mathrm{MAE}_k, \quad \mathrm{QWK}^\star = \max_{k\in\mathcal{M}} \mathrm{QWK}_k,$
$\mathrm{W1}^\star = \max_{k\in\mathcal{M}} \mathrm{W1}_k, \quad \mathrm{TBU}^\star = \min_{k\in\mathcal{M}} \mathrm{TBU}_k$ and let $\delta_R, \delta_A, \delta_Q, \delta_W, \delta_T \geq 0$ be materiality

tolerances.[17] The admissible set of models is then

$$\mathcal{F} = \left\{ m \in \mathcal{M} : \begin{array}{c} \mathrm{RMSE}_m \leq (1+\delta_R)\mathrm{RMSE}^\star \wedge \mathrm{MAE}_m \leq (1+\delta_A)\mathrm{MAE}^\star \\ \wedge\, \mathrm{QWK}_m \geq \mathrm{QWK}^\star - \delta_Q \wedge \mathrm{W1}_m \geq \mathrm{W1}^\star - \delta_W \wedge \mathrm{TBU}_m \leq \mathrm{TBU}^\star + \delta_T \end{array} \right\} \quad (8)$$

and the baseline specification is selected by prioritising distributional credibility within this admissible set:

$$m = \arg\min_{m \in \mathcal{F}} \mathrm{TVD}_m. \quad (9)$$

With only two candidates, this rule is especially transparent: the TVD-leading model is chosen unless it violates one or more guardrails by a material amount. Taken together, the evidence clearly supports *Model I (MultiClass on bins)* as the baseline specification.

The decisive dimension is *distributional credibility in bin space*. Model I achieves an overall TVD of 0.0585 compared with 0.2290 for Model II, with similarly large differences in average TVD by wave, country, and sector. Hence, the regression-plus-snap strategy of Model II materially reshapes the bracket distribution in the holdout period. Given that financing-gap aggregates are computed from these bracket shares, such distortions would mechanically propagate into downstream euro aggregates. In contrast, Model I preserves the empirical bracket composition much more closely while simultaneously delivering stronger ordinal agreement (higher QWK and Macro F1).

[17] Materiality tolerances are chosen to reflect "non-trivial" degradation in secondary objectives: we require euro-space errors to remain within 10% of the best performer ($\delta_R = \delta_A = 0.10$) and allow at most small absolute deteriorations in ordinal and tail diagnostics ($\delta_Q = \delta_W = 0.02$, $\delta_T = 0.05$), so that distributional improvements are not purchased at the cost of meaningful pointwise or tail errors.

Chapter 5

Financing-Gap Estimates: Diagnostics and Descriptive Evidence

This chapter presents descriptive results for the firm-level financing gap constructed using the baseline imputation model selected in Chapter 5 (Model I). Having established the predictive and distributional credibility of the imputed loan-size component, the objective is now to characterise the level, evolution, and heterogeneity of the resulting financing gap across countries, firm sizes, and sectors. Throughout, results are reported for four alternative measurement variants that correspond to distinct *estimands*. Specifically, the variants differ in whether they include (i) only "observed" applicants versus also "shadow" (discouraged) firms, and (ii) the full sample (including zero gaps) versus only firms with strictly positive gaps. Presenting the definitions side by side clarifies how population coverage (observed vs. observed+shadow) and conditioning (unconditional vs. $Gap > 0$) affect both the level and the distribution of the estimated gap.

The analysis remains descriptive in nature. The goal is not to establish causal mechanisms, that is left to future analysis, but to assess the robustness of the methodology, document systematic patterns and structural heterogeneity in the data.

Table 12: Selection of firms under alternative financing-gap definitions

	Observed firms	**Observed and shadow firms**
$Gap \geq 0$	***Method 1*** Considers observations where firms applied for a bank loan (or did not apply because of sufficient internal funds), irrespective of whether the loan demand has been satisfied or not.	***Method 2*** Considers observations where firms applied for a bank loan (or did not apply because of sufficient internal funds) as well as observations where firms have a need but did not apply, irrespective of whether the loan demand has been satisfied or not.
$Gap > 0$	***Method 3*** Considers observations where firms applied for a bank loan, keeping only those where the loan demand has not been satisfied.	***Method 4*** Considers observations where firms applied for a bank loan as well as observations where firms have a need but did not apply, keeping only those where the loan demand has not been satisfied.

5.1 Internal Consistency and Imputation Uncertainty

The comparison in Figure 3 is primarily an internal consistency check of the estimation and imputation pipeline. Two features are informative.

First, the level ordering across methods is stable over time: conditioning on strictly positive gaps (Methods 3-4) shifts the mean upward relative to specifications that include zero-gap firms (Methods 1-2), and adding discouraged ("shadow") firms (Methods 2 and 4) raises the mean relative to observed-only counterparts. This monotonic ranking is implied by the construction of the *NeedFactor* and the sample definitions, and its persistence supports the coherence of the gap measure.

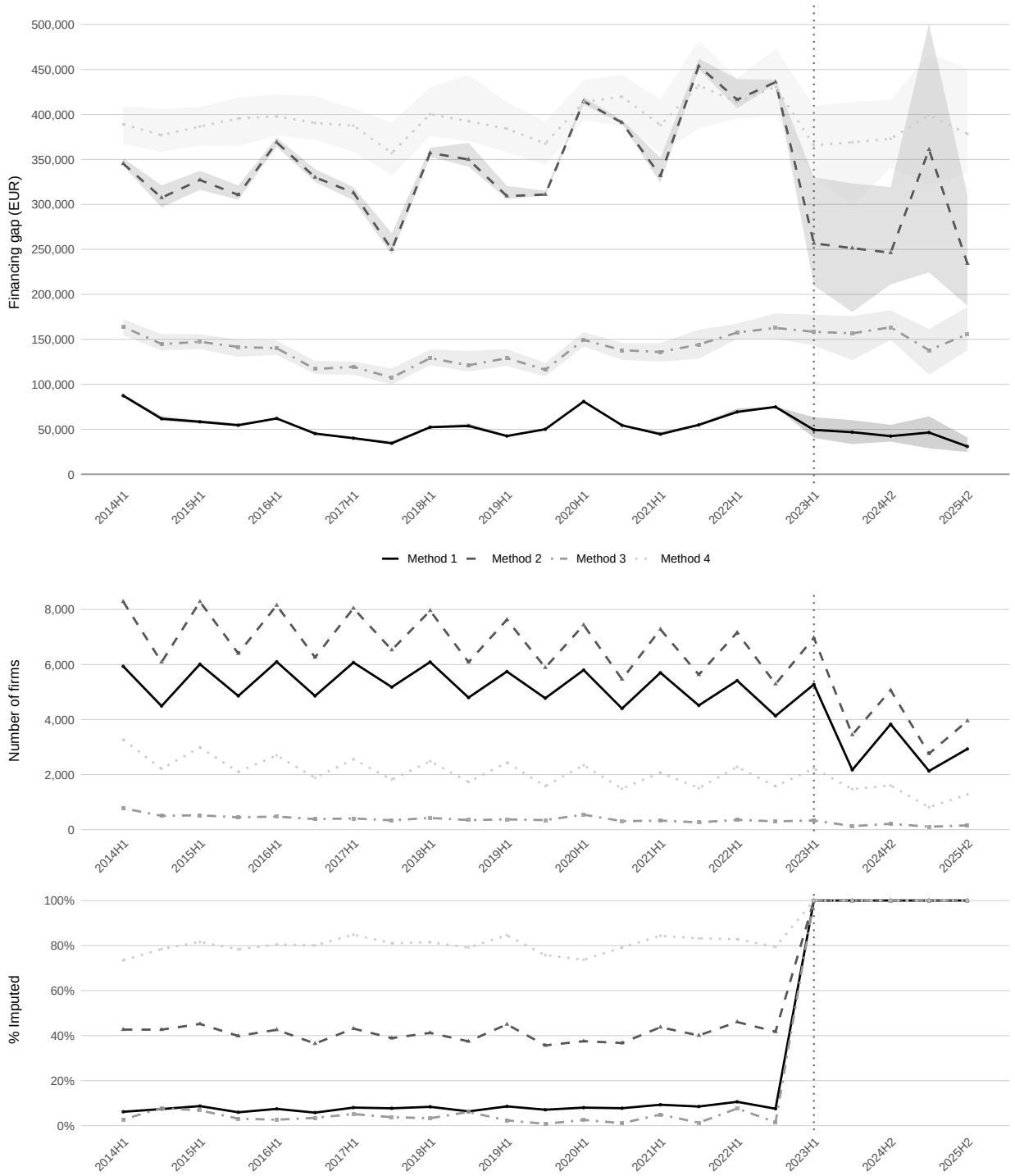
Second, the time profile is broadly synchronised across methods, with common turning points despite large level differences, indicating that the loan-size imputation preserves aggregate time variation rather than generating method-specific volatility. This is particularly salient once the imputation intensity increases sharply: the third panel shows that the share of imputed loan sizes rises markedly (reaching 100% when $Q8A$ becomes structurally unavailable), yet the gap series does not exhibit instability, explosion behaviour or re-ranking, consistent with a well-behaved out-of-time/out-of-sample imputation.

The shaded areas in the top panel represent conditional imputation uncertainty bands. For each firm i with missing $Q8A$, the selected *CatBoost* model provides a predicted categorical distribution $\hat{p}_{ik}$ over loan-size brackets. Missing values are completed by drawing from Categorical$(\hat{p}_i)$ and mapping the draw to its associated midpoint. Holding the estimated model fixed, this stochastic completion is repeated M times, yielding M alternative realisations of the weighted mean gap for each method and semester. The bands correspond to the central 90% interval (5th – 95th percentiles) of this Monte Carlo distribution.

These intervals, therefore, quantify uncertainty arising solely from the probabilistic completion of structurally missing loan-size information. They do not incorporate parameter-estimation or sampling uncertainty. The relatively limited width of the bands – even in periods where imputation becomes universal – indicates that the predictive distributions are sufficiently concentrated to prevent the propagation of imputation noise into aggregate instability.

Finally, the second panel documents the evolution of effective sample sizes across methods. The recurrent oscillatory pattern reflects the semiannual survey design, which mechanically induces seasonality in respondent counts. The pronounced contraction observed from 2024 onward corresponds to a structural change in questionnaire coverage that reduces the number of firms for which the relevant variables are jointly observed. Importantly, these compositional shifts do not induce structural breaks in level ordering or relative dynamics across methods, suggesting that the aggregate gap measures are robust to survey-design perturbations.

Figure 3: Weighted mean firm-level financing gap



Notes: The semester 2023H2 is not shown due to missing data in the original SAFE release. Shaded areas denote 90% conditional imputation uncertainty bands (5th–95th percentiles), obtained from repeated stochastic completion of missing loan-size observations with the estimated model held fixed. The vertical dotted line marks the discontinuity of question $Q8A$ (wave 29). From 2024 onward, the decline in the number of firms reflects a survey-design change rather than a contraction in financing demand.

5.1.1 Cross-method differences

The four specifications differ in *estimands* rather than in statistical disagreement. Let Gap_{it} denote the firm-level financing gap and let $\mathcal{S}^{obs}$ be the set of observed applicant firms, while $\mathcal{S}^{obs+sh}$ augments it with discouraged ("shadow") firms. Methods 1–2 estimate the unconditional weighted mean $\mathbb{E}_w[Gap_{it} \mid i \in \mathcal{S}]$ (including zero-gap firms), whereas Methods 3–4 estimate the conditional mean among strictly positive gaps $\mathbb{E}_w[Gap_{it} \mid Gap_{it} > 0,\ i \in \mathcal{S}]$. Separately, Methods 2 and 4 expand the target population from $\mathcal{S}^{obs}$ to $\mathcal{S}^{obs+sh}$. Hence, the resulting level differences can be understood as *premia*. Table 13 summarises them in a decomposition that isolates the marginal contribution of each modelling choice.

Table 13: Average method premia

Premium (definition)	Full sample	Pre-2020	2020–2022	Post-2022
Shadow premium (unconditional): $Gap^{(2)} - Gap^{(1)}$	97 446	91 487	95 020	124 201
Positive-gap premium (observed): $Gap^{(3)} - Gap^{(1)}$	328 902	302 923	382 456	351 696
Positive-gap premium (with shadow): $Gap^{(4)} - Gap^{(2)}$	316 894	297 033	350 709	335 316
Shadow premium (conditional on $Gap > 0$): $Gap^{(4)} - Gap^{(3)}$	85 438	85 597	63 273	107 820
Net difference (smallest vs largest sample): $Gap^{(3)} - Gap^{(2)}$	231 455	211 436	287 436	227 496

Notes: $Gap^{(m)}$ denotes the weighted mean gap under Method m. The sample is split into three subperiods: pre-2020 (baseline), 2020–2022 (pandemic period under observed $Q8A$), and post-2022 (structural Q8A missingness with full imputation).

The cross-method premia display a temporary increase in 2020-2022, consistent with pandemic-related macro-financial stress, but revert to levels comparable to the pre-2020 period thereafter. Importantly, premia remain stable in the post-2022 full-imputation regime, indicating that the machine-learning recovery of $Q8A$ does not mechanically distort level differences across specifications. This stability supports the methodological credibility of the baseline imputation framework.

5.2 Baseline Aggregate Financing Gaps

Using the baseline specification (Method 4), total financing gaps are computed as

$$TG_{c,t} = \sum_{i \in c,t} w_{it} Gap_{it}, \tag{10}$$

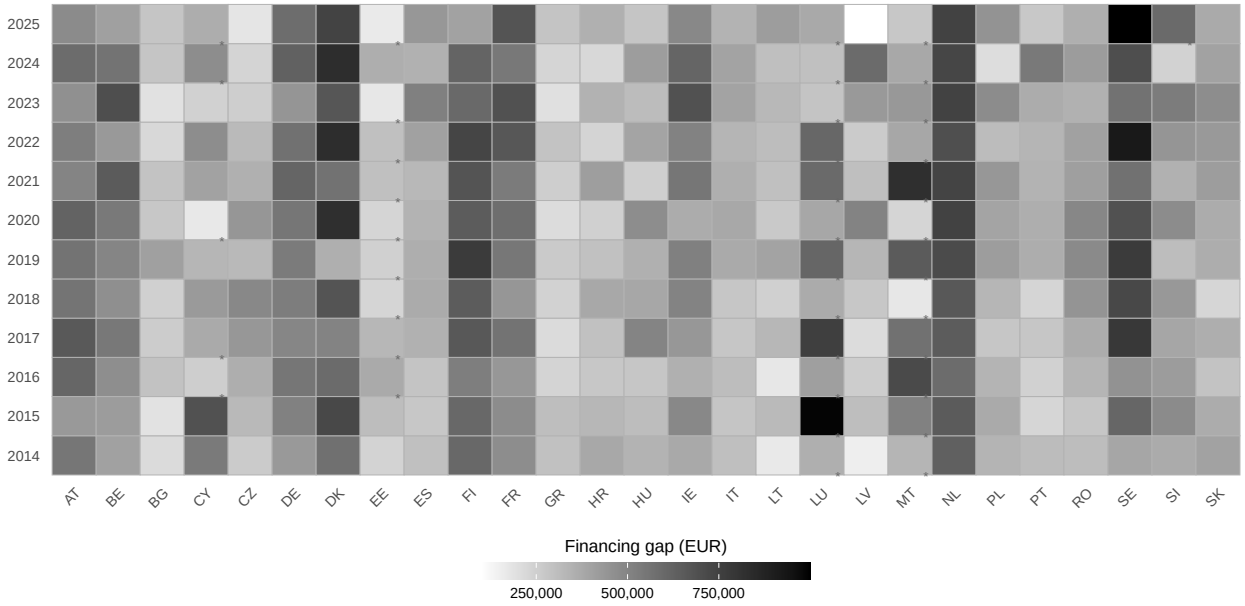
so that aggregates jointly reflect the incidence of positive gaps and their monetary magnitude. They should therefore be interpreted as extensive-intensive composite measures of credit frictions rather than pure demand indicators. In particular, total gaps increase either because a larger number of firms experience unmet financing needs (extensive margin), because the shortfall per constrained firm becomes larger (intensive margin), or both.

The country-year heatmap (Figure 4) shows substantial and persistent cross-country heterogeneity. High-gap countries remain structurally above low-gap ones across multiple waves, including within the euro area, confirming that financing conditions appear to be fragmented despite a common monetary policy. At the same time, cyclical movements – most notably during the pandemic – are broadly synchronised across countries, suggesting that the measure captures genuine macro-financial shocks rather than artificial volatility.

Firm size emerges as a systematic dimension of heterogeneity. The country–size heatmap (Figure 5) reveals a clear monotonic gradient in total financing gaps: large firms are associated with the highest aggregate euro gaps, followed by medium, small, and micro firms. This is

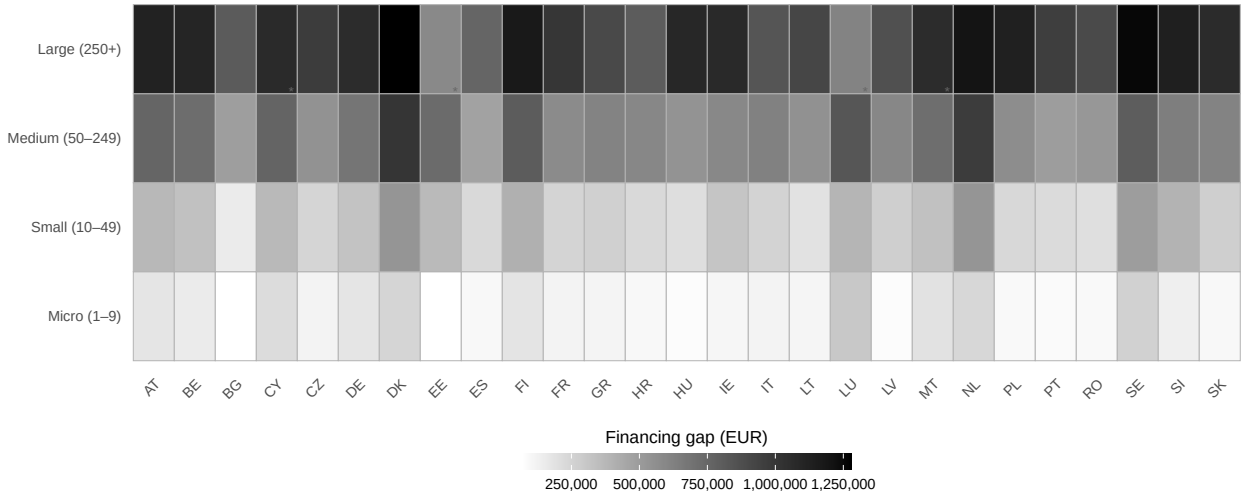
probably due to the larger scale of loan requests among bigger firms rather than a higher incidence of constraints. Sectoral differences are present but comparatively moderate (Figure 6). Overall, the stability of cross-country rankings, the coherent size gradient, and the common cyclical patterns, at these aggregate levels, support the distributional credibility of the baseline imputation model. These aggregates therefore, provide a consistent basis for the subsequent methodological and regression analysis.

Figure 4: Total financing gap by country and year (baseline Method 4)



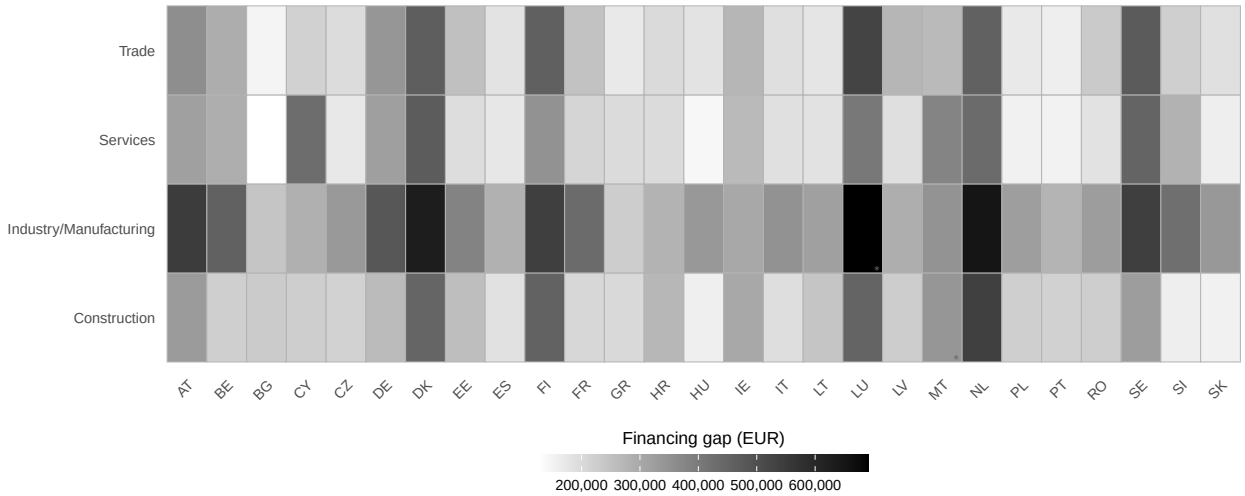
Notes: An asterisk (*) denotes cells based on fewer than 20 observations.

Figure 5: Post-2020 total financing gap by country and size (baseline Method 4)



Notes: An asterisk (*) denotes cells with fewer than 20 observations.

Figure 6: Post-2020 total financing gap by country and sector (baseline Method 4)



Notes: An asterisk (*) denotes cells with fewer than 20 observations.

Chapter 6

Conclusions

This thesis develops a consistent and expandable framework to measure firm-level financing gaps in the EU using SAFE microdata, addressing the structural break created by the discontinuation of $Q8A$ loan-size information after 2022H2. By framing loan-size recovery as an imputation and model-selection problem, and comparing a multiclass and a continuous specification, the analysis shows that model choice materially affects the distributional properties of the recovered loan-size variable and, consequently, aggregate financing-gap measures. Importantly, the imputation relies exclusively on SAFE internal variables, ensuring that the resulting dataset can subsequently be used for regression and policy analysis without *mechanical overlap* between the imputation stage and potential explanatory covariates.

Model selection is governed by an explicit and portable decision rule designed to prioritise credibility over single-score optimisation: bin-space distributional credibility is treated as the binding constraint, euro-space accuracy as a guardrail, and gap-space diagnostics as sanity checks. Under this rule, the multiclass specification (Model I) is selected as the baseline, while the continuous specification remains a robustness benchmark. This approach ensures that downstream aggregates remain interpretable, comparable, and economically coherent.

Using the selected baseline, the thesis constructs financing gaps under alternative measurement variants and documents persistent cross-country heterogeneity, a pronounced size gradi-

ent, and broadly shared cyclical movements around major macro-financial events. The framework therefore restores continuity to euro-valued financing-gap measurement beyond 2022 and provides a coherent basis for comparative and policy-relevant analysis.

Overall, the main contribution of the thesis lies in its transparent, rule-based methodology, which aligns model evaluation with the ultimate object of interest – credible aggregates – rather than maximising isolated predictive metrics. Model selection is governed by the explicit constrained decision rule introduced in Section 4.5, which prioritises distributional credibility in bin space subject to euro-space and tail-risk guardrails. In doing so, the framework delivers a consistent and ready-to-use firm-level dataset that restores euro-valued financing-gap measurement beyond 2022 and provides a solid empirical foundation for policy-relevant and causal analysis. More broadly, the proposed architecture – separating model construction, multi-dimensional validation, and rule-based selection – is expandable and can be applied to other missing-data problems in survey environments, where preserving distributional structure and aggregate coherence is central to empirical credibility.

The principal limitation is that uncertainty remains conditional on the estimated model and does not incorporate sampling or model-estimation variability. Future work could extend the uncertainty assessment (e.g. through bootstrap resampling or repeated re-training), strengthen external validation against administrative credit data, and broaden the modelling framework by testing alternative algorithms and enriching predictors with macroeconomic and institutional variables such as inflation, policy rates, industrial production, unemployment, sovereign spreads, financial conditions indices, and indicators of institutional quality and banking-sector health.

Appendix A

Computational Details and Code

All codes used to clean the SAFE microdata, implement the imputation models, and produce the tables and figures in this thesis is available in a public replication repository: `https://github.com/tommasopetrelli/catboost-SAFE-imputation`. The empirical analysis relies on publicly available SAFE data obtained from `https://www.ecb.europa.eu/stats/ecb_surveys/safe/html/data.en.html`. The dataset was last accessed on 10 February 2026 and reflects the information available at that date. After download, the raw files are imported and processed exclusively using the scripts provided in the replication repository, following the procedures described in Chapters 2 and 3. All cleaning, imputation, and model estimation procedures were performed on a MacBook Air equipped with an Apple M2 processor. For Model I, the total computational time amounted to 21.6 hours, primarily due to the exhaustive grid search conducted for hyperparameter tuning. The optimal configuration selected by the grid search consisted of a tree depth of 6, a learning rate of 0.05, and an L2 regularisation parameter of 3. Model II was estimated under the same computational environment, yielding a total runtime of 1.9 hours and the following optimal hyperparameters: depth = 8, learning rate = 0.05, and $l2_leaf_reg$ = 5.

The data-cleaning pipeline and predictor selection reported below are identical for Model I and Model II. The two approaches diverge only in the imputation stage, namely, the loss function/-

target definition and the associated hyperparameter tuning (grid search) and CatBoost training routines. For brevity, this appendix reports the grid-search and training code for Model I only; Model II is documented in the replication repository.

A.1 Data cleaning

```
# ==============================================================================
# BLOCK 2     Load + clean data
# ==============================================================================

# --- 2.1  Load raw data, aggregate quarterly waves, deduplicate ---------------
#
#   Semiannual regime (waves 1  29  ): wave_agg = wave (unchanged)
#   Quarterly regime  (waves 30+):  consecutive quarter-pairs collapsed into
#                                   sequential semiannual indices:
#     30  31   (2024  Q 1 Q2 )     30 (2024 H1)
#     32  33   (2024  Q 3 Q4 )     31 (2024 H2)
#     34  35   (2025  Q 1 Q2 )     32 (2025 H1)
#     36  37   (2025  Q 3 Q4 )     33 (2025 H2)
#
#   Deduplication: one row per (permid, wave_agg); for the ~980 firms appearing
#   in both quarters of the same half-year, the earlier quarter's record is kept.

dat_agg <- read_dta(path_raw) %>%
  mutate(
    permid    = as.character(permid),
    orig_wave = coerce_wave_to_int(wave),
    wave_agg  = case_when(
      orig_wave %in% c(30L, 31L) ~ 30L,
      orig_wave %in% c(32L, 33L) ~ 31L,
      orig_wave %in% c(34L, 35L) ~ 32L,
      orig_wave %in% c(36L, 37L) ~ 33L,
      TRUE                       ~ orig_wave
    )
  ) %>%
  arrange(permid, wave_agg, orig_wave) %>%
  distinct(permid, wave_agg, .keep_all = TRUE)
```

```r
attributes(dat_agg)[c("label", "datalabel", "notes")] <- NULL
# drop any leftover Stata file-level label if present

# --- 2.2  Apply sample filters + q8a cleaning  --------------------------------
#   Uses dat_agg directly     no additional disk read.
#   Same country/wave/financing filters and q8a recoding as original V5.

dataset_A_raw <- dat_agg %>%
  filter(
    d0 %in% c("BE","DK","DE","EE","IE","GR","ES","FR","HR","IT",
              "CY","LV","LT","LU","MT","NL","AT","PL","PT","RO",
              "SI","SK","FI","SE","BG","CZ","HU"),
    wave_agg > 10,
    q7a_a %in% c(1, 2) | q32 %in% c(1, 2, 3, 4, 5, 6)
  ) %>%
  select(
    -c(wgtcommon, wgtentr, wgtoldentr, wgtoldcommon,
       ida, experiment_1, experiment_2, intdate),
    -any_of("q8a_rec")
  ) %>%
  arrange(permid, wave_agg)

dataset_A <- dataset_A_raw %>%
  mutate(
    across(
      -c(permid, wave_agg),
      ~ if (inherits(.x, "haven_labelled")) haven::as_factor(.x) else as.factor(.x)
    ),
    q8a = as.character(q8a),
    q8a = stringr::str_trim(q8a),
    q8a = dplyr::na_if(q8a, ""),
    q8a = stringr::str_extract(q8a, "\\d+"),
    q8a = ifelse(q8a == "9", NA_character_, q8a),
    q8a = factor(q8a)
  )

cat("2.2␣done␣|␣Rows:", nrow(dataset_A),
    "|␣Columns:", ncol(dataset_A),
    "|␣Missing␣q8a:", sum(is.na(dataset_A$q8a)), "\n")
```

A.2 Predictors selection

```
# ==============================================================================
# BLOCK 3     Feature selection + NA encoding (hybrid rule)
# ==============================================================================
id_vars    <- c("permid", "wave_agg", "d0")
target_var <- "q8a"
leak_vars  <- c("q8a")

hard_drop <- names(which(colMeans(is.na(dataset_A)) > 0.90))
dataset_A <- dataset_A %>% select(-all_of(hard_drop))

candidate_vars <- setdiff(names(dataset_A), c(id_vars, leak_vars))
vars_to_keep   <- candidate_vars[
  sapply(candidate_vars, function(v) missingness_informative(dataset_A, v, target_var))
]
vars_to_keep <- union(vars_to_keep, intersect("q7a_a", names(dataset_A)))

dataset_A <- dataset_A %>%
  select(all_of(c(id_vars, target_var, vars_to_keep)))

pred_cols <- setdiff(names(dataset_A), c(id_vars, leak_vars))
dataset_A <- dataset_A %>%
  mutate(across(all_of(pred_cols),
                ~ if (is.factor(.x)) forcats::fct_explicit_na(.x, "MISSING") else .x))

cat("Step␣3␣completed␣|␣Columns␣after␣missingness␣control:", ncol(dataset_A), "\n")
```

A.3 Grid Search

```
# ==============================================================================
# BLOCK 6     One-shot grid search on holdout waves (diagnostic only)
# ==============================================================================
grid <- expand.grid(
  depth         = c(5, 6, 8),
  learning_rate = c(0.03, 0.05),
  l2_leaf_reg   = c(3, 5)
)
```

```r
results <- vector("list", nrow(grid))

for (i in seq_len(nrow(grid))) {
  params <- list(
    loss_function       = "MultiClass",
    eval_metric         = "MultiClass",
    iterations          = 3000L,
    learning_rate       = grid$learning_rate[i],
    depth               = grid$depth[i],
    l2_leaf_reg         = grid$l2_leaf_reg[i],
    od_type             = "Iter",
    od_wait             = 100,
    random_seed         = 123,
    thread_count        = n_cores,
    use_best_model      = TRUE,
    logging_level       = "Silent",
    allow_writing_files = FALSE
  )

  m         <- catboost.train(train_pool, test_pool, params = params)
  probs_i   <- catboost.predict(m, test_pool, prediction_type = "Probability")
  logloss_i <- safe_logloss(probs_i, y_test_cls, n_classes)

  best_iter_i <- tryCatch(catboost.get_best_iteration(m) + 1L,
                          error = function(e) params$iterations)

  results[[i]] <- list(params = grid[i, ], best_iter = best_iter_i, logloss = logloss_i)

  cat(sprintf("Grid␣%02d/%02d␣|␣Logloss␣%.5f␣|␣Iter␣%d␣|␣depth=%d␣lr=%.3f␣l2=%d\n",
              i, nrow(grid), logloss_i, best_iter_i,
              grid$depth[i], grid$learning_rate[i], grid$l2_leaf_reg[i]))

  rm(m, probs_i); gc()
}

best_id <- which.min(sapply(results, `[[`, "logloss"))
best    <- results[[best_id]]

cat("\nBest␣grid␣choice:\n"); print(best$params)
cat(sprintf("Best␣holdout␣logloss:␣%.5f␣|␣Best␣iteration:␣%d\n", best$logloss, best$best_iter))
```

A.4 Catboost model

```
# ==============================================================================
# BLOCK 7     Final model (fit on ALL observed q8a)
# ==============================================================================
freq_all         <- table(y_testable_cls)
w_map_all        <- median(as.numeric(freq_all)) / as.numeric(freq_all)
names(w_map_all) <- names(freq_all)
w_all            <- as.numeric(w_map_all[as.character(y_testable_cls)])
w_all[is.na(w_all)] <- 1

all_pool <- catboost.load_pool(
  X_all, y_testable_cls,
  cat_features = cat_features, weight = w_all
)

final_params <- list(
  loss_function  = "MultiClass",
  eval_metric    = "MultiClass",
  iterations     = best$best_iter,
  learning_rate  = best$params$learning_rate,
  depth          = best$params$depth,
  l2_leaf_reg    = best$params$l2_leaf_reg,
  random_seed    = 123,
  thread_count   = n_cores,
  logging_level  = "Silent"
)

final_model <- catboost.train(all_pool, params = final_params)
```

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AI-based tools were used in a supporting role in the preparation of this thesis. All analytical choices, methodological decisions, interpretations, and conclusions presented in this thesis remain solely my own responsibility.